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<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-
scale=1.0, user-scalable=no"/>
<meta name="apple-mobile-web-app-capable" content="yes"/>
<meta name="apple-mobile-web-app-status-bar-style" content="black-translucent"/>
<meta name="apple-mobile-web-app-title" content="Tyvek by Mike"/>
<meta name="theme-color" content="#001022"/>
<title>Tyvek by Mike</title>
<link rel="preconnect" href="https://fonts.googleapis.com"/>
<link href="https://fonts.googleapis.com/css2?
family=Sora:wght@300;400;600;700;800;900&family=JetBrains+Mono:wght@400;700&display
rel="stylesheet"/>
<style>
:root{--navy:#001022;--navy2:#001d3d;--blue:#0057b8;--accent:#00b4d8;--
red:#e63946;--orange:#f77f00;--yellow:#f4d03f;--green:#06d6a0;--gray:#94a3b8;--
radius:16px;--shadow:0 4px 24px rgba(0,16,34,0.13);}
*{box-sizing:border-box;margin:0;padding:0;-webkit-tap-highlight-
color:transparent;}
html,body{height:100%;font-family:'Sora',sans-serif;background:var(--
navy);color:var(--navy);overflow:hidden;}
body{display:flex;flex-direction:column;}
#header{background:linear-gradient(135deg,var(--navy) 0%,var(--navy2)
100%);padding:env(safe-area-inset-top,16px) 16px 0;padding-top:calc(env(safe-area-
inset-top,0px) + 14px);border-bottom:1px solid rgba(0,180,216,0.15);flex-
shrink:0;position:relative;overflow:hidden;}
#header::before{content:'';position:absolute;top:-40px;right:-40px;width:160px;heig
gradient(circle,rgba(0,180,216,0.12),transparent 70%);border-radius:50%;}
.header-inner{display:flex;align-items:center;justify-content:space-
between;position:relative;}
.logo-wrap{display:flex;align-items:center;gap:10px;}
.logo-icon{width:38px;height:38px;border-radius:11px;background:linear-
gradient(135deg,var(--accent),var(--blue));display:flex;align-
items:center;justify-content:center;font-size:20px;box-shadow:0 4px 12px
rgba(0,180,216,0.35);}
.logo-text{color:#fff;}
.logo-title{font-size:17px;font-weight:800;letter-spacing:-0.3px;line-height:1;}
.logo-sub{font-size:9px;font-weight:600;letter-spacing:2px;text-
transform:uppercase;color:var(--accent);margin-top:2px;}
.header-score{text-align:right;}
.score-num{font-size:22px;font-weight:900;line-height:1;color:var(--green);}
.score-lbl{font-size:9px;color:var(--gray);font-weight:600;text-

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transform:uppercase;letter-spacing:1px;}
.tabs{display:flex;gap:0;margin-top:12px;}
.tab{flex:1;padding:10px 4px;text-align:center;font-size:11px;font-
weight:700;color:var(--gray);letter-spacing:0.5px;text-
transform:uppercase;cursor:pointer;border-bottom:2px solid
transparent;transition:all 0.2s;}
.tab.active{color:var(--accent);border-bottom-color:var(--accent);}
#main{flex:1;overflow-y:auto;overflow-x:hidden;background:#f0f4fa;-webkit-
overflow-scrolling:touch;}
.view{display:none;padding:16px;min-height:100%;}
.view.active{display:block;animation:fadeUp 0.25s ease;}
@keyframes
fadeUp{from{opacity:0;transform:translateY(8px);}to{opacity:1;transform:translateY(
@keyframes spin{to{transform:rotate(360deg);}}}
.card{background:#fff;border-radius:var(--radius);box-shadow:var(--
shadow);overflow:hidden;margin-bottom:14px;}
.card-header{background:linear-gradient(135deg,var(--navy),var(--
navy2));padding:12px 16px;display:flex;align-items:center;justify-content:space-
between;}
.card-title{color:#fff;font-size:12px;font-weight:700;text-
transform:uppercase;letter-spacing:1px;}
.card-body{padding:14px 16px;}
.field-grid{display:grid;grid-template-columns:1fr 1fr;gap:10px;}
.field{display:flex;flex-direction:column;gap:4px;}
.field.full{grid-column:1/-1;}
.field label{font-size:10px;font-weight:700;color:var(--gray);text-
transform:uppercase;letter-spacing:0.8px;}
.field input,.field select,.field textarea{border:1.5px solid #e2e8f0;border-
radius:9px;padding:9px 12px;font-size:13px;font-family:'Sora',sans-
serif;color:var(--navy);background:#fafbff;outline:none;transition:border-color
0.2s;-webkit-appearance:none;}
.field input:focus,.field select:focus{border-color:var(--accent);}
.field textarea{resize:vertical;min-height:64px;line-height:1.5;}
.add-btn{width:100%;background:linear-gradient(135deg,var(--navy),var(--
blue));border:2px dashed rgba(0,180,216,0.4);border-radius:var(--
radius);padding:22px 16px;display:flex;flex-direction:column;align-
items:center;gap:7px;cursor:pointer;margin-bottom:16px;transition:all 0.2s;box-
shadow:0 6px 20px rgba(0,87,184,0.2);}
.add-btn:active{transform:scale(0.98);}
.add-btn-icon{font-size:32px;}
.add-btn-title{color:#fff;font-size:15px;font-weight:800;}
.add-btn-sub{color:var(--accent);font-size:11px;font-weight:600;}
.photo-card{background:#fff;border-radius:var(--radius);overflow:hidden;margin-
bottom:16px;box-shadow:var(--shadow);}
.photo-img-wrap{position:relative;}
.photo-img{width:100%;height:240px;object-fit:cover;display:block;}
.photo-badges{position:absolute;top:10px;left:10px;display:flex;flex-

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wrap:wrap;gap:5px;}
.badge{font-size:10px;font-weight:700;padding:3px 9px;border-
radius:20px;background:rgba(0,16,34,0.78);color:#fff;backdrop-
filter:blur(4px);letter-spacing:0.3px;}
.status-badge{position:absolute;top:10px;right:10px;font-size:11px;font-
weight:800;padding:4px 10px;border-radius:20px;color:#fff;}
.delete-
btn{position:absolute;bottom:10px;right:10px;background:rgba(230,57,70,0.88);border
radius:50%;cursor:pointer;font-size:16px;display:flex;align-items:center;justify-
content:center;}
.analyzing-
overlay{position:absolute;inset:0;background:rgba(0,16,34,0.82);display:flex;flex-
direction:column;align-items:center;justify-content:center;gap:12px;}
.spinner{width:40px;height:40px;border:3px solid rgba(255,255,255,0.15);border-
top-color:var(--accent);border-radius:50%;animation:spin 0.8s linear infinite;}
.analyzing-text{color:#fff;font-size:13px;font-weight:700;}
.analysis-body{padding:16px;}
.compliance-row{display:flex;justify-content:space-between;align-
items:center;margin-bottom:6px;}
.compliance-label{font-size:11px;font-weight:700;color:var(--gray);text-
transform:uppercase;letter-spacing:0.5px;}
.bar-bg{background:#eef1f7;border-radius:8px;height:8px;overflow:hidden;margin-
bottom:14px;}
.bar-fill{height:100%;border-radius:8px;}
.section-title{font-size:11px;font-weight:800;color:var(--navy);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:8px;}
.problem-item{border-radius:11px;padding:11px 13px;margin-bottom:8px;}
.problem-title{font-size:13px;font-weight:700;color:var(--navy);line-
height:1.35;flex:1;}
.problem-ref{font-size:11px;color:var(--gray);font-style:italic;margin-
bottom:5px;}
.problem-solution{font-size:12px;color:#2563eb;font-weight:600;line-height:1.5;}
.pass-box{background:#f0fff8;border-radius:11px;padding:10px 13px;border:1px solid
rgba(6,214,160,0.2);margin-bottom:10px;}
.note-box{background:#f8faff;border-left:3px solid var(--accent);border-radius:0
10px 10px 0;padding:10px 13px;margin-top:10px;}
.note-label{font-size:10px;font-weight:700;color:var(--accent);text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:3px;}
.note-text{font-size:12px;color:#445;line-height:1.5;}
.photo-meta{padding:10px 16px;border-top:1px solid #eef1f7;display:flex;justify-
content:space-between;align-items:center;}
.meta-time{font-size:11px;color:#aaa;font-family:'JetBrains Mono',monospace;}
.meta-note{font-size:11px;color:#778;max-width:55%;overflow:hidden;white-
space:nowrap;text-overflow:ellipsis;}
.empty{text-align:center;padding:48px 20px;}
.empty-icon{font-size:52px;margin-bottom:12px;}
.empty-title{font-size:16px;font-weight:700;color:#445;margin-bottom:6px;}

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.empty-sub{font-size:13px;color:#aaa;}
.report-summary{background:linear-gradient(135deg,var(--navy),var(--navy2));border-radius:var(--radius);padding:20px;margin-bottom:16px;box-shadow:0 8px 28px rgba(0,16,34,0.2);}
.report-project{color:var(--accent);font-size:10px;font-weight:700;text-transform:uppercase;letter-spacing:2px;margin-bottom:4px;}
.report-name{color:#fff;font-size:20px;font-weight:900;line-height:1.2;margin-bottom:2px;}
.report-address{color:var(--gray);font-size:12px;margin-bottom:16px;}
.stats-grid{display:grid;grid-template-columns:repeat(4,1fr);gap:8px;}
.stat-box{background:rgba(255,255,255,0.08);border-radius:10px;padding:10px 6px;text-align:center;}
.stat-val{font-size:20px;font-weight:900;line-height:1;}
.stat-lbl{font-size:9px;color:var(--gray);font-weight:600;text-transform:uppercase;margin-top:2px;}
.gen-btn{width:100%;background:linear-gradient(135deg,#0057b8,#0091c2);color:#fff;border:none;border-radius:var(--radius);padding:16px;font-size:15px;font-weight:800;font-family:'Sora',sans-serif;cursor:pointer;box-shadow:0 6px 20px rgba(0,87,184,0.3);transition:all 0.2s;margin-bottom:14px;}
.gen-btn:active{transform:scale(0.98);}
.gen-btn:disabled{opacity:0.5;cursor:not-allowed;}
.modal-bg{position:fixed;inset:0;background:rgba(0,16,34,0.88);z-index:500;display:flex;align-items:flex-end;justify-content:center;}
.modal-sheet{background:#fff;border-radius:20px 20px 0 0;width:100%;max-width:600px;padding:20px 16px calc(env(safe-area-inset-bottom,16px) + 16px);max-height:92vh;overflow-y:auto;animation:slideUp 0.3s ease;}
@keyframes slideUp{from{transform:translateY(100%);}to{transform:translateY(0);}}
.modal-handle{width:40px;height:4px;background:#e2e8f0;border-radius:2px;margin:auto 16px;}
.modal-title{font-size:16px;font-weight:800;color:var(--navy);margin-bottom:14px;}
.modal-preview{width:100%;height:180px;object-fit:cover;border-radius:12px;margin-bottom:14px;}
.modal-btn{width:100%;background:linear-gradient(135deg,var(--navy),var(--navy2));color:#fff;border:none;border-radius:12px;padding:15px;font-size:14px;font-weight:800;font-family:'Sora',sans-serif;cursor:pointer;}
.cancel-btn{width:100%;background:#f3f4f6;color:#445;border:none;border-radius:12px;padding:13px;font-size:13px;font-weight:700;font-family:'Sora',sans-serif;cursor:pointer;margin-top:8px;}
.toast{position:fixed;top:calc(env(safe-area-inset-top,0px) + 60px);left:50%;transform:translateX(-50%);background:var(--navy);color:#fff;padding:10px 20px;border-radius:20px;font-size:13px;font-weight:600;z-index:999;white-space:nowrap;animation:toastIn 0.3s ease,toastOut 0.3s ease 2.2s forwards;}
@keyframes toastIn{from{opacity:0;transform:translateX(-50%) translateY(-8px);}to{opacity:1;transform:translateX(-50%) translateY(0);}}
@keyframes toastOut{to{opacity:0;}}

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.install-banner{background:linear-gradient(135deg,#003580,#0057b8);border-
radius:13px;padding:14px;margin-bottom:14px;display:flex;align-
items:center;gap:12px;}
.install-icon{font-size:24px;flex-shrink:0;}
.install-text{flex:1;}
.install-title{color:#fff;font-size:13px;font-weight:700;margin-bottom:2px;}
.install-sub{color:rgba(255,255,255,0.65);font-size:11px;line-height:1.4;}
.install-close{color:rgba(255,255,255,0.5);font-size:20px;cursor:pointer;flex-
shrink:0;}
@media print{body{overflow:visible;background:#fff;}#header,.no-
print{display:none!important;}#main{overflow:visible;}.view{display:block!important
inspect,#view-project,#view-report{display:none!important;}#print-
target{display:block!important;}}
</style>
</head>
<body>

<div id="header">
  <div class="header-inner">
    <div class="logo-wrap">
      <div class="logo-icon"><img alt="Tyvek logo icon" data-bbox="382 428 402 443"/></div>
      <div class="logo-text">
        <div class="logo-title">Tyvek by Mike</div>
        <div class="logo-sub">DuPont AI Inspector</div>
      </div>
    </div>
    <div class="header-score" id="header-score" style="display:none">
      <div class="score-num" id="avg-score">-</div>
      <div class="score-lbl">Compliance</div>
    </div>
  </div>
  <div class="tabs">
    <div class="tab active" onclick="switchTab('inspect')"><img alt="Inspection icon" data-bbox="668 660 688 675"/> Observation</div>
    <div class="tab" onclick="switchTab('project')"><img alt="Project icon" data-bbox="600 680 620 695"/> Project</div>
    <div class="tab" onclick="switchTab('report')"><img alt="Report icon" data-bbox="592 698 612 713"/> Report</div>
  </div>
</div>

<div id="main">

  <!-- INSPECT -->
  <div class="view active" id="view-inspect">
    <div class="install-banner" id="install-banner">
      <div class="install-icon"><img alt="Install icon" data-bbox="418 868 438 883"/></div>
      <div class="install-text">
        <div class="install-title">Install on iPhone</div>
        <div class="install-sub">Tap <strong>Share</strong> → <strong>Add to Home

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Screen</strong></div>
    </div>
    <div class="install-close" onclick="document.getElementById('install-
banner').style.display='none'">×</div>
    </div>
    <button class="add-btn" onclick="document.getElementById('file-
input').click()">
        <div class="add-btn-icon">📷</div>
        <div class="add-btn-title">Take / Upload Photo</div>
        <div class="add-btn-sub">AI detects elevation · floor · issues ·
solutions</div>
    </button>
    <input type="file" id="file-input" accept="image/*" capture="environment"
style="display:none" onchange="handleFile(event)"/>
    <div id="photos-container">
        <div class="empty" id="empty-state">
            <div class="empty-icon">📷</div>
            <div class="empty-title">No photos yet</div>
            <div class="empty-sub">Add photos of the Tyvek installation<br>to begin
your observation</div>
        </div>
    </div>
</div>

<!-- PROJECT -->
<div class="view" id="view-project">
    <div class="card">
        <div class="card-header"><span class="card-title">📁 Project
Information</span></div>
        <div class="card-body">
            <div class="field-grid">
                <div class="field full"><label>Project Name</label><input id="proj-name"
placeholder="e.g. Las Palmas Residential"/></div>
                <div class="field full"><label>Address</label><input id="proj-addr"
placeholder="e.g. 123 Main St, Miami FL 33101"/></div>
                <div class="field"><label>Inspector</label><input id="proj-inspector"
placeholder="Your name"/></div>
                <div class="field"><label>Company</label><input id="proj-company"
placeholder="Your company"/></div>
                <div class="field"><label>Date</label><input type="date" id="proj-
date"/></div>
                <div class="field"><label>Phone</label><input type="tel" id="proj-phone"
placeholder="+1 (305) 000-0000"/></div>
                <div class="field full"><label>Email</label><input type="email"
id="proj-email" placeholder="mike@company.com"/></div>
                <div class="field full"><label>Project Notes</label><textarea id="proj-
notes" placeholder="General description, site conditions, weather, etc.">

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</textarea></div>
    </div>
</div>
</div>
<div class="card">
    <div class="card-header"><span class="card-title"><img alt="Building icon" data-bbox="638 141 658 156"/> Building Photo (cover
page)</span></div>
    <div class="card-body">
        <button class="add-btn" style="margin-bottom:0;padding:16px"
onclick="document.getElementById('building-input').click()">
            <div class="add-btn-icon" id="building-icon"><img alt="Building icon" data-bbox="628 238 648 253"/></div>
            <div class="add-btn-title" id="building-label">Upload building
photo</div>
            <div class="add-btn-sub">Will appear on the report cover page</div>
        </button>
        <input type="file" id="building-input" accept="image/*"
style="display:none" onchange="handleBuilding(event)"/>
        <img id="building-preview" style="display:none;width:100%;border-
radius:10px;margin-top:12px;max-height:200px;object-fit:cover;"/>
    </div>
</div>
</div>

<!-- REPORT -->
<div class="view" id="view-report">
    <div class="report-summary">
        <div class="report-project">Project</div>
        <div class="report-name" id="rpt-name">Unnamed Project</div>
        <div class="report-address" id="rpt-addr">No address provided</div>
        <div class="stats-grid">
            <div class="stat-box"><div class="stat-val" id="rpt-total"
style="color:var(--accent)">0</div><div class="stat-lbl">Photos</div></div>
            <div class="stat-box"><div class="stat-val" id="rpt-critical"
style="color:var(--red)">0</div><div class="stat-lbl">Critical</div></div>
            <div class="stat-box"><div class="stat-val" id="rpt-major"
style="color:var(--orange)">0</div><div class="stat-lbl">Major</div></div>
            <div class="stat-box"><div class="stat-val" id="rpt-score"
style="color:var(--green)">-</div><div class="stat-lbl">Score</div></div>
        </div>
    </div>
    <button class="gen-btn" id="gen-btn" onclick="generateReport()"><img alt="Generate icon" data-bbox="751 813 771 828"/> Generate
Professional Report</button>
    <div id="report-output"></div>
</div>

</div>

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<!-- MODAL -->
<div class="modal-bg" id="modal" style="display:none">
  <div class="modal-sheet">
    <div class="modal-handle"></div>
    <div class="modal-title">📷 Observation Photo Details</div>
    <img id="modal-preview" class="modal-preview" src="" alt=""/>
    <div class="field-grid" style="margin-bottom:14px">
      <div class="field"><label>Floor / Level</label><input id="m-floor"
placeholder="e.g. Floor 2, Level 3"/></div>
      <div class="field"><label>Elevation</label>
        <select id="m-elev">
          <option>North</option><option>South</option><option>East</option>
<option>West</option>
          <option>Northeast</option><option>Northwest</option>
<option>Southeast</option><option>Southwest</option>
          <option>Interior</option><option>Exterior</option><option>Roof</option>
<option>Other</option>
        </select>
      </div>
      <div class="field full"><label>Status</label>
        <select id="m-status">
          <option>Under Observation</option><option>In Progress</option>
<option>Completed</option>
          <option>Needs Review</option><option>Approved</option>
<option>Rejected</option>
        </select>
      </div>
      <div class="field full"><label>Additional Note</label><input id="m-note"
placeholder="Quick observation (optional)"/></div>
    </div>
    <button class="modal-btn" onclick="confirmPhoto()">🧐 Analyze with Tyvek
AI</button>
    <button class="cancel-btn" onclick="cancelPhoto()">Cancel</button>
  </div>
</div>

<div id="print-target" style="display:none;"></div>

<script>
let photos=[], pendingDataUrl=null, buildingDataUrl=null;

const TYVEK_PROMPT=`You are a senior construction inspector and DuPont Tyvek
certified specialist based in the United States.
Analyze the construction photo for weather barrier (Tyvek/housewrap) installation
compliance.

DuPont Tyvek Key Guidelines:

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- Horizontal overlaps: minimum 6 inches, upper over lower like shingles
- Vertical overlaps: minimum 6 inches, taped with DuPont Tyvek Tape
- Fastening: cap staples/nails every 18in along studs, 12in at edges/openings
- All seams must be taped with DuPont Tyvek Tape or approved flashing tape
- Max UV exposure: 120 days (HomeWrap), 4 months (CommercialWrap)
- Window/door openings: sill pan flashing required before WRB integration
- Penetrations (pipes, outlets, vents): sealed with flexible flashing tape min 6in overlap
- No installation over wet or dirty substrate
- Tears/punctures: patch with Tyvek tape min 6 inches beyond damage in all directions
- Bottom termination: termination bar or proper tape seal required
- Wrinkles over 1/2 inch must be corrected (moisture trap risk)
- Corners: use Tyvek tape or pre-formed corner pieces

Look for floor number clues (signage, floor count, window patterns) and elevation/direction clues (sun angle, street view, compass, shadows).

Respond ONLY with this exact JSON (no markdown, no extra text):

```
{"overall_status":"CRITICAL|MAJOR|MINOR|PASS","compliance_score":0,"detected_floor":1,"id":1,"severity":"CRITICAL|MAJOR|MINOR","problem":"","location_detail":"","guide":[],"inspector_notes":"","immediate_action_required":false}`;
```

```
const statusColor=s=>
({CRITICAL:'#e63946',MAJOR:'#f77f00',MINOR:'#f4d03f',PASS:'#06d6a0'}
[s]||'#94a3b8');
const statusBg=s=>
({CRITICAL:'#fff0f0',MAJOR:'#fff8ee',MINOR:'#ffdf0',PASS:'#f0fff8'}
[s]||'#f8faff');
const sevIcon=s=>({CRITICAL:'🔴',MAJOR:'🟠',MINOR:'🟡',PASS:'🟢'}[s]||'⬜');

function showToast(msg)
{document.querySelectorAll('.toast').forEach(t=>t.remove());const
t=document.createElement('div');t.className='toast';t.textContent=msg;document.body

function switchTab(name){
  document.querySelectorAll('.tab').forEach((t,i)=>t.classList.toggle('active',
['inspect','project','report'][i]===name));
  document.querySelectorAll('.view').forEach(v=>v.classList.remove('active'));
  document.getElementById('view-'+name).classList.add('active');
  if(name==='report') updateReportSummary();
}

function handleFile(e){
  const file=e.target.files[0];if(!file)return;
  const r=new FileReader();r.onload=ev=>
{pendingDataUrl=ev.target.result;openModal();};r.readAsDataURL(file);e.target.value
```

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}
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```
function handleBuilding(e) {
  const file=e.target.files[0];if(!file)return;
  const r=new FileReader();r.onload=ev=>{
    buildingDataUrl=ev.target.result;
    document.getElementById('building-preview').src=buildingDataUrl;
    document.getElementById('building-preview').style.display='block';
    document.getElementById('building-label').textContent='Photo loaded ✓';
    document.getElementById('building-icon').textContent='✅';
  };r.readAsDataURL(file);e.target.value='';
}
```

```
function openModal(){document.getElementById('modal-
preview').src=pendingDataUrl;document.getElementById('m-
floor').value='';document.getElementById('modal').style.display='flex';}
function cancelPhoto()
{document.getElementById('modal').style.display='none';pendingDataUrl=null;}
```

```
async function confirmPhoto(){
  const floor=document.getElementById('m-floor').value;
  const elev=document.getElementById('m-elev').value;
  const status=document.getElementById('m-status').value;
  const note=document.getElementById('m-note').value;
  document.getElementById('modal').style.display='none';
  const photo=
{id:Date.now(),dataUrl:pendingDataUrl,floor,elevation:elev,status(note,timestamp:ne
Date().toISOString(),analyzing:true,analysis:null);
  photos.push(photo);pendingDataUrl=null;
  renderPhoto(photo);
  document.getElementById('empty-state').style.display='none';
  try{
    const analysis=await analyzePhoto(photo.dataUrl);
    photo.analyzing=false;photo.analysis=analysis;
    if(!photo.floor&&analysis.detected_floor)photo.floor=analysis.detected_floor;

if(analysis.detected_elevation&&photo.elevation==='North')photo.elevation=analysis.
  updatePhotoCard(photo);updateHeaderScore();
  showToast(analysis.overall_status==='PASS'?'✅ No issues detected':'!⚠️
Analysis complete');
  }catch(err){
    photo.analyzing=false;
    photo.analysis={overall_status:'MINOR',compliance_score:50,problems:
[{'id:1,severity:'MINOR',problem:'Analysis error. Check your internet
connection.',guideline_reference:'N/A',solution:'Retry with a stable internet
connection.',location_detail:''}],passing_items:[],inspector_notes:'AI connection
error.',tyvek_visible:false,installation_stage:'Unknown',detected_floor:null,detect
```

```

        updatePhotoCard(photo);
    }
}

async function analyzePhoto(dataUrl) {
    const base64=dataUrl.split(',')[1];
    const res=await fetch('https://api.anthropic.com/v1/messages',
{method:'POST',headers:{'Content-
Type':'application/json'},body:JSON.stringify({model:'claude-sonnet-4-
20250514',max_tokens:1000,system:TYVEK_PROMPT,messages:[{role:'user',content:
[{type:'image',source:{type:'base64',media_type:'image/jpeg',data:base64}},
{type:'text',text:'Analyze this construction photo for Tyvek installation
compliance. Return ONLY JSON.'}]}})});
    const data=await res.json();
    const text=data.content?.find(b=>b.type==='text')?.text||'{}';
    return JSON.parse(text.replace(/``json|``/g,'').trim());
}

function renderPhoto(photo) {
    const c=document.getElementById('photos-container');
    const d=document.createElement('div');d.className='photo-card';d.id='photo-
'+photo.id;d.innerHTML=photoCardHTML(photo);c.appendChild(d);
}

function updatePhotoCard(photo){const d=document.getElementById('photo-
'+photo.id);if(d)d.innerHTML=photoCardHTML(photo);}

function photoCardHTML(photo) {
    const a=photo.analysis;const sc=a?statusColor(a.overall_status):'#94a3b8';
    let body='';
    if(a&&!photo.analyzing) {
        const
scoreC=a.compliance_score>=80?'#06d6a0':a.compliance_score>=60?'#f77f00': '#e63946';
        body=`<div class="analysis-body">
            <div class="compliance-row"><span class="compliance-label">DuPont Tyvek
Compliance</span><span style="font-size:14px;font-
weight:900;color:${scoreC}">${a.compliance_score}%</span></div>
            <div class="bar-bg"><div class="bar-fill"
style="width:${a.compliance_score}%;background:${scoreC}"></div></div>
            ${a.installation_stage?`<div style="font-size:12px;color:#556;padding:8px
11px;background:#f0f4fa;border-radius:9px;margin-bottom:10px;">🔍
${a.installation_stage}</div>`:''}
            ${a.problems&&a.problems.length?`<div class="section-title">⚠ Issues Found
(${a.problems.length})</div>${a.problems.map(p=>`<div class="problem-item"
style="background:${statusBg(p.severity)};border:1px solid
${statusColor(p.severity)}22"><div style="display:flex;gap:7px;align-items:flex-
start;margin-bottom:5px"><span>${sevIcon(p.severity)}</span><span class="problem-
title">${p.problem}</span></div>${p.location_detail?`<div class="problem-ref">📍

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```

    ${p.location_detail}</div>`:'')<div class="problem-ref"><img alt="book icon" data-bbox="638 48 658 60"/>
    ${p.guideline_reference}</div><div class="problem-solution"><img alt="checkmark icon" data-bbox="673 65 693 77"/> SOLUTION:
    ${p.solution}</div></div>`).join('')}`:'')
        ${a.passing_items}&&a.passing_items.length?`<div class="pass-box"><div
class="section-title" style="color:#047857;margin-bottom:4px"><img alt="checkmark icon" data-bbox="773 123 793 135"/> Correct
Installation</div>${a.passing_items.map(i=>`<div style="font-
size:12px;color:#047857;line-height:1.6;">• ${i}</div>`).join('')}</div>`:'')
        ${a.inspector_notes?`<div class="note-box"><div class="note-label"><img alt="notepad icon" data-bbox="798 183 818 195"/> AI
Inspector Note</div><div class="note-text">${a.inspector_notes}</div></div>`:'')
        </div>`;
    }
    return `<div class="photo-img-wrap">
        
        <div class="photo-badges">${photo.floor?`<span class="badge"><img alt="floor icon" data-bbox="723 298 743 310"/> ${photo.floor}
</span>`:'')<span class="badge"><img alt="elevation icon" data-bbox="413 315 433 327"/> ${photo.elevation}</span><span class="badge"><img alt="status icon" data-bbox="863 315 883 327"/>
${photo.status}</span></div>
        ${a?`<div class="status-badge"
style="background:${sc}">${sevIcon(a.overall_status)} ${a.overall_status}
</div>`:'')
        ${photo.analyzing?`<div class="analyzing-overlay"><div class="spinner"></div>
<div class="analyzing-text">Analyzing with Tyvek AI...</div></div>`:'')
        <button class="delete-btn" onclick="deletePhoto(${photo.id})">×</button>
        </div>${body}
        <div class="photo-meta"><span class="meta-time">${new
Date(photo.timestamp).toLocaleString('en-US',
{dateStyle:'short',timeStyle:'short'})}</span>${photo.note?`<span class="meta-
note"><img alt="pencil icon" data-bbox="173 548 193 560"/> ${photo.note}</span>`:'')}</div>`;
    }

function deletePhoto(id){photos=photos.filter(p=>p.id!==id);const
el=document.getElementById('photo-
'+id);if(el)el.remove();if(!photos.length)document.getElementById('empty-
state').style.display='block';updateHeaderScore();}

function updateHeaderScore(){
    const a=photos.filter(p=>p.analysis);
    if(!a.length){document.getElementById('header-
score').style.display='none';return;}
    const avg=Math.round(a.reduce((s,p)=>s+p.analysis.compliance_score,0)/a.length);
    document.getElementById('header-score').style.display='block';
    document.getElementById('avg-score').textContent=avg+'%';
    document.getElementById('avg-
score').style.color=avg>=80?`#06d6a0`:avg>=60?`#f77f00`:`#e63946`;
}

function updateReportSummary(){
    document.getElementById('rpt-name').textContent=document.getElementById('proj-

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name').value||'Unnamed Project';
    document.getElementById('rpt-addr').textContent=document.getElementById('proj-
addr').value||'No address provided';
    document.getElementById('rpt-total').textContent=photos.length;
    const a=photos.filter(p=>p.analysis);
    document.getElementById('rpt-
critical').textContent=a.filter(p=>p.analysis.overall_status==='CRITICAL').length;
    document.getElementById('rpt-
major').textContent=a.filter(p=>p.analysis.overall_status==='MAJOR').length;
    const avg=a.length?
Math.round(a.reduce((s,p)=>s+p.analysis.compliance_score,0)/a.length):null;
    document.getElementById('rpt-score').textContent=avg!==null?avg+'%':'-';
}

function generateReport(){
    if(!photos.length){showToast('⚠️ Add photos first');return;}
    const btn=document.getElementById('gen-
btn');btn.disabled=true;btn.textContent='⌚ Generating...';
    const n=v=>document.getElementById(v).value;
    const projName=n('proj-name')||'Unnamed Project';
    const projAddr=n('proj-addr')||'Address not specified';
    const inspector=n('proj-inspector')||'Inspector';
    const company=n('proj-company')||'';
    const phone=n('proj-phone')||'';
    const email=n('proj-email')||'';
    const projDate=n('proj-date')||new Date().toLocaleDateString('en-US');
    const projNotes=n('proj-notes')||'';
    const analyzed=photos.filter(p=>p.analysis);
    const avgScore=analyzed.length?
Math.round(analyzed.reduce((s,p)=>s+p.analysis.compliance_score,0)/analyzed.length)
    const allProblems=analyzed.flatMap(p=>(p.analysis.problems||[]).map(pr=>
({...pr,photoFloor:p.floor,photoElev:p.elevation})));
    const criticalProblems=allProblems.filter(p=>p.severity==='CRITICAL');
    const crits=analyzed.filter(p=>p.analysis.overall_status==='CRITICAL').length;
    const majs=analyzed.filter(p=>p.analysis.overall_status==='MAJOR').length;
    const scoreC=avgScore>=80?'#06d6a0':avgScore>=60?'#f77f00':'#e63946';
    const verdict=avgScore>=80?'APPROVED':avgScore>=60?'CONDITIONAL':'REJECTED';
    const vColor=avgScore>=80?'#06d6a0':avgScore>=60?'#f77f00':'#e63946';

    const html=`<div id="rpt-doc" style="font-family:'Sora',sans-
serif;background:#fff;border-radius:16px;overflow:hidden;box-shadow:0 8px 40px
rgba(0,16,34,0.15);">
    <div style="background:linear-gradient(160deg,#001022 0%,#001d3d 60%,#003580
100%);padding:40px 28px 32px;position:relative;overflow:hidden;">
        <div
style="position:absolute;top:-60px;right:-60px;width:240px;height:240px;background:
gradient(circle,rgba(0,180,216,0.15),transparent 70%);border-radius:50%;"></div>

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<div style="display:flex;align-items:center;gap:10px;margin-bottom:28px;">
  <div style="width:40px;height:40px;background:linear-
gradient(135deg,#00b4d8,#0057b8);border-radius:11px;display:flex;align-
items:center;justify-content:center;font-size:20px;">🏠</div>
  <div><div style="color:#fff;font-size:16px;font-weight:800;">Tyvek by
Mike</div><div style="color:#00b4d8;font-size:9px;font-weight:700;letter-
spacing:2px;text-transform:uppercase;">DuPont AI Inspector</div></div>
  <div style="margin-left:auto;background:${vColor};color:#fff;padding:5px
14px;border-radius:20px;font-size:12px;font-weight:800;">${verdict}</div>
</div>
  ${buildingDataUrl?``:''}
  <div style="color:#00b4d8;font-size:10px;font-weight:700;letter-
spacing:2px;text-transform:uppercase;margin-bottom:6px;">Construction Observation
Report</div>
  <div style="color:#fff;font-size:24px;font-weight:900;line-height:1.15;margin-
bottom:6px;">${projName}</div>
  <div style="color:#94a3b8;font-size:13px;margin-bottom:20px;">📍 ${projAddr}
</div>
  <div style="display:grid;grid-template-columns:1fr 1fr;gap:10px;margin-
bottom:20px;">
    <div style="background:rgba(255,255,255,0.07);border-
radius:10px;padding:12px;">
      <div style="color:#94a3b8;font-size:9px;text-transform:uppercase;letter-
spacing:1px;margin-bottom:3px;">Inspector</div>
      <div style="color:#fff;font-size:13px;font-weight:700;">${inspector}</div>
      ${company?`<div style="color:#94a3b8;font-size:11px;">${company}
</div>`:''}
      ${email?`<div style="color:#94a3b8;font-size:11px;">${email}</div>`:''}
    </div>
    <div style="background:rgba(255,255,255,0.07);border-
radius:10px;padding:12px;">
      <div style="color:#94a3b8;font-size:9px;text-transform:uppercase;letter-
spacing:1px;margin-bottom:3px;">Observation Date</div>
      <div style="color:#fff;font-size:13px;font-weight:700;">${projDate}</div>
      ${phone?`<div style="color:#94a3b8;font-size:11px;">${phone}</div>`:''}
      <div style="color:#94a3b8;font-size:11px;">Generated: ${new
Date().toLocaleDateString('en-US',{dateStyle:'long'})}</div>
    </div>
  </div>
  <div style="display:grid;grid-template-columns:repeat(4,1fr);gap:8px;">
    ${[ {v:photos.length,l:'Photos',c:'#00b4d8'},
{v:crits,l:'Critical',c:'#e63946'}, {v:maj,s,l:'Major',c:'#f77f00'},
{v:avgScore+'%',l:'Score',c:scoreC} ].map(s=>`<div
style="background:rgba(255,255,255,0.08);border-radius:10px;padding:10px 6px;text-
align:center;"><div style="font-size:20px;font-weight:900;color:${s.c};line-

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height:1;">${s.v}</div><div style="font-size:9px;color:#94a3b8;font-
weight:600;text-transform:uppercase;margin-top:2px;">${s.l}</div>
</div>`).join('')
    </div>
</div>

<div style="padding:24px;">
    <div style="font-size:11px;font-weight:800;color:#001022;text-
transform:uppercase;letter-spacing:1px;margin-bottom:12px;padding-
bottom:6px;border-bottom:2px solid #eef1f7;"><img alt="Executive Summary icon" data-bbox="535 218 555 235"/> Executive Summary</div>
    <div style="background:linear-gradient(135deg,#f8faff,#eef4ff);border-
radius:13px;padding:16px;margin-bottom:16px;border-left:4px solid ${vColor};">
        <div style="font-size:12px;font-weight:700;color:${vColor};margin-
bottom:6px;">OVERALL VERDICT: ${verdict}</div>
        <div style="font-size:13px;color:#334;line-height:1.6;">The observation of
<strong>${projName}</strong> recorded an average compliance score of <strong
style="color:${scoreC}">${avgScore}%</strong> against official DuPont Tyvek
installation guidelines. A total of <strong>${allProblems.length}</strong> finding(s)
</strong> were documented across ${photos.length} photo(s).
${criticalProblems.length?`<strong
style="color:#e63946">${criticalProblems.length}</strong> CRITICAL issue(s)</strong>
require immediate corrective action.`:'No critical issues were detected.'}</div>
        </div>
        ${projNotes?`<div style="background:#fffbf0;border-
radius:12px;padding:14px;margin-bottom:16px;border:1px solid #ffc30022;"><div
style="font-size:10px;font-weight:700;color:#b8941a;text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:4px;">Project Notes</div>
<div style="font-size:12px;color:#445;line-height:1.6;">${projNotes}</div>
</div>`:''}
        ${criticalProblems.length?`<div style="background:#fff0f0;border-
radius:12px;padding:14px;margin-bottom:16px;border:1px solid #e6394622;"><div
style="font-size:10px;font-weight:700;color:#e63946;text-
transform:uppercase;letter-spacing:0.8px;margin-bottom:8px;"><img alt="Immediate Actions Required icon" data-bbox="685 658 705 675"/> Immediate Actions
Required</div>${criticalProblems.map(p=>`<div style="font-
size:12px;color:#7b1d1d;margin-bottom:6px;padding:7px
10px;background:rgba(230,57,70,0.08);border-radius:8px;line-height:1.4;">
<strong>${p.problem}</strong>${p.photoFloor?` - '+p.photoFloor+'`}
(${p.photoElev||''})<div style="color:#e63946;font-size:11px;margin-top:3px;">→
${p.solution}</div></div>`).join('')}</div>`:''}
    </div>

<div style="padding:0 24px 24px;">
    <div style="font-size:11px;font-weight:800;color:#001022;text-
transform:uppercase;letter-spacing:1px;margin-bottom:12px;padding-
bottom:6px;border-bottom:2px solid #eef1f7;"><img alt="Compliance by Photo icon" data-bbox="535 888 555 905"/> Compliance by Photo</div>
    ${analyzed.map((p,i)=>{const sc=p.analysis.compliance_score;const
scC=sc>=80?'#06d6a0':sc>=60?'#f77f00':'#e63946';return`<div

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style="display:flex;align-items:center;gap:10px;margin-bottom:8px;"><div
style="font-size:11px;color:#667;min-width:72px;font-family:'JetBrains
Mono',monospace;">Photo ${i+1}${p.floor?' ' +p.floor.substring(0,6):''}</div><div
style="flex:1;background:#eef1f7;border-radius:6px;height:14px;overflow:hidden;">
<div style="width:${sc}%;height:100%;background:${scC};border-radius:6px;"></div>
</div><div style="font-size:12px;font-weight:800;color:${scC};min-width:36px;text-
align:right;">${sc}%</div><div style="font-size:10px;font-weight:700;padding:2px
8px;border-
radius:10px;background:${scC}22;color:${scC};">${p.analysis.overall_status}</div>
</div>`;)).join('')
</div>

<div style="padding:0 24px 24px;">
  <div style="font-size:11px;font-weight:800;color:#001022;text-
transform:uppercase;letter-spacing:1px;margin-bottom:16px;padding-
bottom:6px;border-bottom:2px solid #eef1f7;">📷 Observation Photo Details</div>
  ${photos.map((photo,i)=>{
    const a=photo.analysis;const sc=a.statusColor(a.overall_status):'#94a3b8';
    return`<div style="border:1.5px solid ${sc}33;border-
radius:14px;overflow:hidden;margin-bottom:20px;background:#fff;">
      <div style="background:${sc};padding:10px 14px;display:flex;align-
items:center;justify-content:space-between;">
        <div style="color:#fff;font-size:13px;font-weight:800;">Photo
        ${i+1}${photo.floor?' ' +photo.floor:''}</div>
        <div style="display:flex;gap:6px;"><span
style="background:rgba(255,255,255,0.2);color:#fff;padding:2px 9px;border-
radius:10px;font-size:10px;font-weight:700;">📏 ${photo.elevation}</span><span
style="background:rgba(255,255,255,0.2);color:#fff;padding:2px 9px;border-
radius:10px;font-size:10px;font-weight:700;">${photo.status}</span></div>
        </div>
        
        <div style="padding:14px;">
          ${a?`<div style="display:flex;justify-content:space-between;align-
items:center;margin-bottom:8px;"><span style="font-size:11px;font-
weight:700;color:#667;text-transform:uppercase;letter-spacing:0.5px;">Tyvek
Compliance</span><span style="font-size:16px;font-
weight:900;color:${a.compliance_score}>=80?'#06d6a0':a.compliance_score>=60?'#f77f00
</span></div>
          <div style="background:#eef1f7;border-
radius:6px;height:8px;overflow:hidden;margin-bottom:12px;"><div
style="width:${a.compliance_score}%;height:100%;background:${a.compliance_score}>=80
radius:6px;"></div></div>
          ${a.installation_stage?`<div style="font-
size:12px;color:#556;padding:8px 11px;background:#f0f4fa;border-radius:8px;margin-
bottom:10px;">🔍 ${a.installation_stage}</div>`:''}
          ${a.problems&&a.problems.length?`<div style="font-size:10px;font-

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weight:800;color:#001022;text-transform:uppercase;letter-spacing:0.8px;margin-
bottom:7px;">⚠ ISSUES FOUND (${a.problems.length})</div>${a.problems.map(p=>`<div
style="background:${statusBg(p.severity)};border:1px solid
${statusColor(p.severity)}33;border-radius:10px;padding:10px 12px;margin-
bottom:7px;"><div style="display:flex;gap:6px;align-items:flex-start;margin-
bottom:4px;"><span>${sevIcon(p.severity)}</span><strong style="font-
size:12px;color:#1a2440;line-height:1.35;flex:1;">${p.problem}</strong>
</div>${p.location_detail?`<div style="font-size:11px;color:#889;font-
style:italic;margin-bottom:3px;">📍 ${p.location_detail}</div>`:''}<div
style="font-size:11px;color:#889;font-style:italic;margin-bottom:5px;">📖
${p.guideline_reference}</div><div style="font-size:12px;color:#1d4ed8;font-
weight:600;line-height:1.4;padding:6px 9px;background:rgba(37,99,235,0.06);border-
radius:7px;">✅ <strong>SOLUTION:</strong> ${p.solution}</div>
</div>`).join('')}`:``}

    ${a.passing_items&&a.passing_items.length?`<div
style="background:#f0fff8;border-radius:10px;padding:10px 12px;margin-
bottom:8px;border:1px solid rgba(6,214,160,0.2)"><div style="font-size:10px;font-
weight:800;color:#047857;text-transform:uppercase;margin-bottom:5px;">✓ CORRECT
INSTALLATION</div>${a.passing_items.map(it=>`<div style="font-
size:12px;color:#065f46;line-height:1.7;">• ${it}</div>`).join('')}</div>`:``}

    ${a.inspector_notes?`<div style="background:#f8faff;border-left:3px
solid #00b4d8;border-radius:0 9px 9px 0;padding:10px 12px;"><div style="font-
size:10px;font-weight:700;color:#00b4d8;text-transform:uppercase;letter-
spacing:0.8px;margin-bottom:3px;">AI INSPECTOR NOTE</div><div style="font-
size:12px;color:#334;line-height:1.5;">${a.inspector_notes}</div></div>`:``}

    `:``<div style="font-size:13px;color:#aaa;padding:10px 0;">No analysis
available</div>`}

    <div style="margin-top:10px;font-size:10px;color:#bbb;font-
family:'JetBrains Mono',monospace;">${new
Date(photo.timestamp).toLocaleString('en-US',
{dateStyle:'short',timeStyle:'short'})}</div>

    </div>
  </div>`;
  }).join('')
</div>

  <div style="background:linear-gradient(135deg,#001022,#001d3d);padding:20px
24px;display:flex;justify-content:space-between;align-items:center;">
    <div><div style="color:#fff;font-size:12px;font-weight:700;">Tyvek by Mike
    DuPont AI Inspector</div><div style="color:#94a3b8;font-size:10px;margin-
top:2px;">Based on official DuPont Tyvek installation guidelines</div></div>
    <div style="text-align:right;"><div style="color:#00b4d8;font-size:18px;font-
weight:900;">${avgScore}%</div><div style="color:#94a3b8;font-size:9px;text-
transform:uppercase;letter-spacing:1px;">Compliance</div></div>
  </div>

  <div style="padding:16px 24px;" class="no-print">
    <button onclick="window.print()" style="width:100%;background:linear-

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gradient(135deg,#001022,#003580);color:#fff;border:none;border-
radius:12px;padding:15px;font-size:14px;font-weight:800;font-family:'Sora',sans-
serif;cursor:pointer;box-shadow:0 6px 20px rgba(0,16,34,0.25);"> Print / Save as
PDF</button>
</div>
</div>`;

document.getElementById('report-output').innerHTML=html;
btn.disabled=false;btn.textContent='📄 Generate Professional Report';
setTimeout(()=>document.getElementById('rpt-
doc').scrollIntoView({behavior:'smooth'}),100);
}

document.getElementById('proj-date').value=new Date().toISOString().split('T')[0];
if(window.navigator.standalone)document.getElementById('install-
banner').style.display='none';
</script>
</body>
</html>

```